

14)Develop a Student Result Analysis System

HTML form, CSS table styling, JavaScript arrays,objects, functions, conditional statements, loops,Math.max() and Math.min()).

CODE

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Student Result Analysis System</title>
```

```
<style>
```

```
  * {
```

```
    box-sizing: border-box;
```

```
  }
```

```
  body {
```

```
    margin: 0;
```

```
    font-family: Arial, sans-serif;
```

```
    background: linear-gradient(135deg, #667eea, #764ba2);
```

```
    min-height: 100vh;
```

```
    padding: 30px;
```

```
  }
```

```
  .container {
```

```
    max-width: 950px;
```

```
margin: auto;
background: white;
padding: 25px;
border-radius: 15px;
box-shadow: 0 10px 30px rgba(0,0,0,0.25);
}
```

```
h1 {
  text-align: center;
  color: #4b3f72;
  margin-bottom: 25px;
}
```

```
form {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 15px;
  margin-bottom: 25px;
}
```

```
input {
  padding: 12px;
  border: 2px solid #ddd;
  border-radius: 8px;
  font-size: 15px;
}
```

```
button {
```

```
padding: 12px;
border: none;
border-radius: 8px;
background: #4b3f72;
color: white;
font-size: 16px;
cursor: pointer;
}
```

```
button:hover {
    background: #372c5c;
}
```

```
.table-container {
    overflow-x: auto;
}
```

```
table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 20px;
}
```

```
th, td {
    padding: 12px;
    text-align: center;
    border: 1px solid #ddd;
}
```

```
th {  
  background: #4b3f72;  
  color: white;  
}
```

```
tr:nth-child(even) {  
  background: #f4f4f4;  
}
```

```
.analysis {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  gap: 15px;  
  margin-top: 25px;  
}
```

```
.card {  
  padding: 18px;  
  border-radius: 10px;  
  text-align: center;  
  background: #eee;  
}
```

```
.card h3 {  
  margin: 5px;  
  color: #4b3f72;  
}
```

```
.card p {
    font-size: 22px;
    font-weight: bold;
}

@media (max-width: 700px) {
    form, .analysis {
        grid-template-columns: 1fr;
    }
}

</style>
</head>

<body>

<div class="container">

    <h1>Student Result Analysis System</h1>

    <form id="resultForm">

        <input type="text" id="name" placeholder="Student Name" required>

        <input type="number" id="math" placeholder="Maths Mark" min="0" max="100"
required>

        <input type="number" id="science" placeholder="Science Mark" min="0" max="100"
required>

        <input type="number" id="english" placeholder="English Mark" min="0" max="100"
required>

        <button type="submit">Add Student</button>
```

</form>

<div class="table-container">

<table>

<thead>

<tr>

<th>Name</th>

<th>Maths</th>

<th>Science</th>

<th>English</th>

<th>Total</th>

<th>Average</th>

<th>Result</th>

</tr>

</thead>

<tbody id="resultTable"></tbody>

</table>

</div>

<div class="analysis">

<div class="card">

<h3>Highest Mark</h3>

<p id="highest">0</p>

</div>

<div class="card">

<h3>Lowest Mark</h3>

```
<p id="lowest">0</p>
```

```
</div>
```

```
<div class="card">
```

```
<h3>Class Average</h3>
```

```
<p id="classAverage">0</p>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<script>
```

```
let students = [];
```

```
document.getElementById("resultForm").addEventListener("submit", function(event) {  
    event.preventDefault();
```

```
let student = {  
    name: document.getElementById("name").value,  
    math: Number(document.getElementById("math").value),  
    science: Number(document.getElementById("science").value),  
    english: Number(document.getElementById("english").value)  
};
```

```
students.push(student);
```

```
displayResults();
```

```
updateAnalysis();
```

```

    this.reset();
  });

function displayResults() {
  let table = document.getElementById("resultTable");
  table.innerHTML = "";

  students.forEach(function(student) {
    let total = student.math + student.science + student.english;
    let average = total / 3;
    let result = average >= 40 ? "Pass" : "Fail";

    let row = `
      <tr>
        <td>${student.name}</td>
        <td>${student.math}</td>
        <td>${student.science}</td>
        <td>${student.english}</td>
        <td>${total}</td>
        <td>${average.toFixed(2)}</td>
        <td>${result}</td>
      </tr>
    `;

    table.innerHTML += row;
  });
}

```



```
function updateAnalysis() {  
    let marks = [];  
  
    students.forEach(function(student) {  
        marks.push(student.math);  
        marks.push(student.science);  
        marks.push(student.english);  
    });  
  
    if (marks.length > 0) {  
        let highest = Math.max(...marks);  
        let lowest = Math.min(...marks);  
  
        let totalMarks = marks.reduce(function(sum, mark) {  
            return sum + mark;  
        }, 0);  
  
        let average = totalMarks / marks.length;  
  
        document.getElementById("highest").innerText = highest;  
        document.getElementById("lowest").innerText = lowest;  
        document.getElementById("classAverage").innerText = average.toFixed(2);  
    }  
}  
  
</script>  
  
</body>
```

</html>

OUTPUT

Student Result Analysis System

Name	Maths	Science	English	Total	Average	Result
Maria Maxi Milla M	80	93	95	268	89.33	Pass
Savi	83	90	95	268	89.33	Pass

Highest Mark
95

Lowest Mark
80

Class Average
89.33